

VANSH KUMAR

Cybersecurity Professional | Digital Forensics Specialist | SOC Analyst | Penetration Tester

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PROFESSIONAL SUMMARY

Results-driven Cybersecurity Professional with hands-on expertise in Digital Forensics, Threat Intelligence, VAPT, and SOC operations. Demonstrated ability to independently build production-grade security tools — including a full-stack SIEM platform (AegisX) and a Law Enforcement-grade CDR Analysis system. Practical experience with Delhi Police Special Cell (IFSO/N-DISC) in cyber forensics and cybercrime investigation workflows. Certified by industry-recognized programs and actively deepening expertise in Digital Forensics, Malware Analysis, and Incident Response. Passionate about bridging offensive and defensive security to deliver real-world investigative impact.

WORK EXPERIENCE

Cybersecurity & Digital Forensics Intern

IFSO / N-DISC, Special Cell – Delhi Police, Dwarka, New Delhi

April 2026 – May 2026

- Worked within Delhi Police's elite IFSO/N-DISC unit gaining exposure to live cybercrime investigations, digital evidence handling, and forensic workflows
- Designed and delivered an AI-powered **Citizen Guidance Chatbot** — deployed for public-facing use within the unit as a capstone project
- Assisted in CDR/IPDR analysis, tower dump investigations, and structured cyber forensics case documentation
- Acquired practical knowledge of cybercrime investigation SOPs, legal compliance, and chain-of-custody procedures under ACP Prem Chandra Khanduri
- Collaborated with law enforcement officers and cybercrime analysts on active investigation workflows

Malware Analysis Intern (Assignment)

Cyber Cell – Delhi Police

2025

- Conducted static and dynamic malware analysis, extracting IOCs including file hashes, registry keys, and network artifacts
- Performed packet-level inspection using Wireshark to identify malicious C2 communication patterns
- Compiled structured technical reports submitted to law enforcement for investigative use

Cyber Volunteer

UP Police Cybercrime Division

September 2025 – November 2025

- Monitored and reported cybercrime incidents including online fraud, phishing, and social engineering cases
- Assisted in digital evidence documentation and reporting aligned with legal standards

PROJECTS

AegisX SIEM — Intelligent Threat Detection & Response Platform

Python (FastAPI) | PostgreSQL | React | Three.js | Tailwind CSS | JWT | Docker | Nginx

- Architected a **full-stack enterprise SIEM** from scratch supporting Windows Event Logs, Linux Syslogs, Firewall Logs, and Web Server Logs with a unified JSON normalization engine and scalable PostgreSQL schema (JSONB, monthly partitioning)
- Built a **Rule-Based Correlation Engine** detecting brute-force attacks, privilege escalation, account lockouts, and suspicious IP patterns with real-time alerting
- Integrated a **Threat Intelligence module** with IOC matching, malicious IP detection, domain reputation scoring, and file hash verification (MD5/SHA1/SHA256)
- Implemented a **Digital Forensics module** with event timeline reconstruction, user activity tracking, evidence preservation, and court-ready PDF/CSV export

- Delivered a real-time **SOC Dashboard** with 3D globe threat map (Three.js), severity monitoring, and alert management via 50+ REST API endpoints
- Containerized deployment using **Docker Compose + Nginx** with RBAC (Admin/Analyst/Viewer), JWT auth, and full audit logging

CDR Analysis Platform — Law Enforcement Intelligence Tool

Python (FastAPI) | SQLite | HTML/CSS/JavaScript | D3.js | Leaflet.js | Chart.js | ReportLab | JWT

- Designed and built an **offline CDR Analysis platform** purpose-built for Indian Law Enforcement, running on localhost with no external dependencies
- Engineered a **CDR Import Engine** with auto-column detection for CSV/XLSX formats, optimized to handle millions of records via efficient SQLite indexing
- Developed advanced investigative modules: **Network Link Analysis** (D3.js), Tower Movement Maps (Leaflet.js), hour-wise heatmaps, and multi-suspect comparison
- Implemented **AI-assisted intelligence features**: Burner Phone Detection (0–100 scoring), IMEI/IMSI tracking, Gang/Cluster Detection, and Unknown Number Profiler
- Automated **court-ready PDF and Excel report generation** with auto Key Findings, Alibi Checker, Cross-Case Search, and full audit logging for chain-of-custody compliance

Advent of Cyber 2025 — TryHackMe

Completed 24 security challenges across web exploitation, network analysis, OSINT, and incident response

TECHNICAL SKILLS

Offensive Security

VAPT | Web App Pentesting | Network Pentesting | SQL Injection | XSS | Burp Suite | Metasploit | Nmap | SQLmap | Nikto | Hydra | Kali Linux

Defensive Security / SOC

SIEM Development & Operations | Log Analysis | Incident Response | Threat Intelligence | IOC Matching | Network Traffic Monitoring | Splunk

Networking

TCP/IP | HTTP/HTTPS | DNS | Network Protocols | Firewall Log Analysis

Digital Forensics

CDR/IPDR Analysis | Malware Analysis (Static & Dynamic) | IOC Extraction | Evidence Preservation | Timeline Reconstruction | Wireshark

Development

Python (FastAPI) | React | JavaScript | D3.js | Three.js | Leaflet.js | REST API | PostgreSQL | SQLite | SQLAlchemy | Docker | JWT | Nginx

Investigative / OSINT

OSINT Techniques | Link Analysis | CDR/Tower Analysis | Dark Web Investigation | Cybercrime Documentation

EDUCATION

Bachelor of Computer Applications (BCA)

Indira Gandhi National Open University (IGNOU) | 2025 – Ongoing

CERTIFICATIONS

- ◆ Certified Cyber Criminologist (CCC) – Cyber Virtual Labs (Ongoing)
- ◆ Deloitte Cyber Job Simulation – Forage (2025)
- ◆ TATA Cybersecurity Analyst Job Simulation – Forage (2025)
- ◆ Advent of Cyber 2025 – TryHackMe
- ◆ Ethical Hacking Training – Global Institute of Cybersecurity & Ethical Hacking
- ◆ Introduction to Cyber Security & Cybercrime – Simplilearn

KEY HIGHLIGHTS

- Independently built production-grade security tools (SIEM + CDR platform) — demonstrating full-stack engineering and cybersecurity depth
- Internship at Delhi Police's elite IFSO/N-DISC Special Cell — one of India's most specialized cybercrime and digital forensics units
- Practical law enforcement exposure across UP Police Cyber Division and Delhi Police Cyber Cell
- 500+ LinkedIn connections in the Cybersecurity domain — active community contributor
- Actively pursuing Digital Forensics specialization with focus on CDR/IPDR analysis, malware investigation, and forensic tooling

References available upon request • Open to On-site, Hybrid & Remote opportunities across India